

Mini Diagnostic Tool for Cars

Presenter Paweł Ćwikowski
278150



ASSUMPTIONS & GOALS

ASSUMPTION: Creating a proprietary, bidirectional OBD-II diagnostic scanner from scratch. This device leverages an ESP32 running FreeRTOS to efficiently filter and decode high-speed CAN bus telemetry in real-time. Furthermore, it moves beyond passive data logging by actively sending protocol commands to request live metrics and clear Diagnostic Trouble Codes directly from the Engine Control Unit.

INSPIRATION:



Tested CARS

Renault Scenic	2008
Renault Traffic	2008
Peugeot 508	2016
Volkswagen Polo	2011

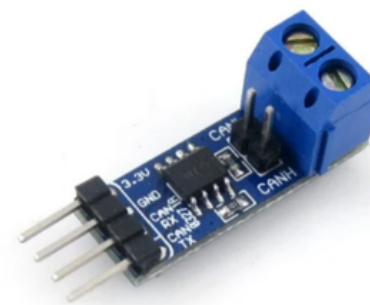
Expected Results

- Succesful Communication with Car
- Implementation of queries for specific PIDs and displaying them on OLED

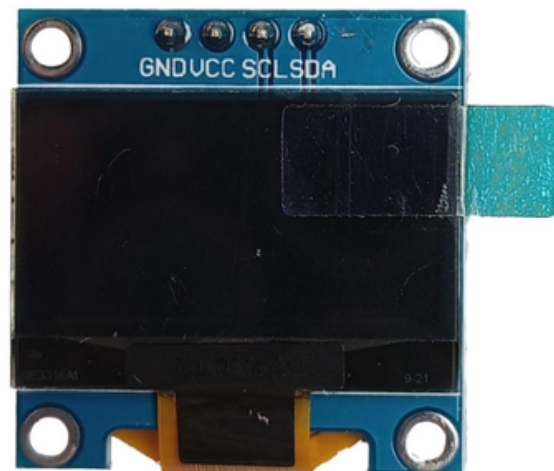
HARDWARE SETUP



- ESP32 DevKit



3V3 CAN Transreciever Module –



- Oled Display



OBD – II Female Cable -

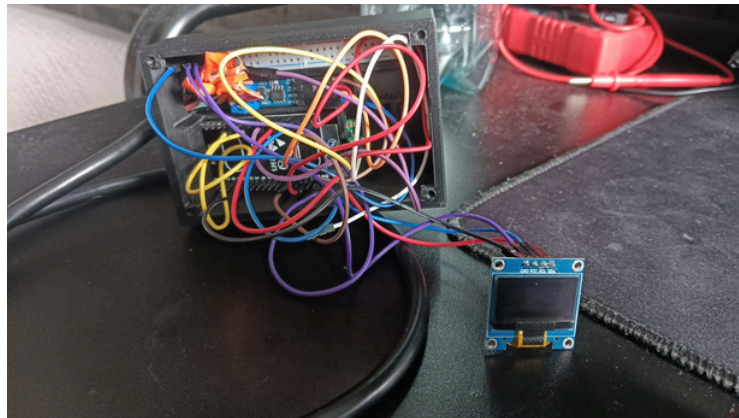
Pololu Buck Converter 24V – 5V



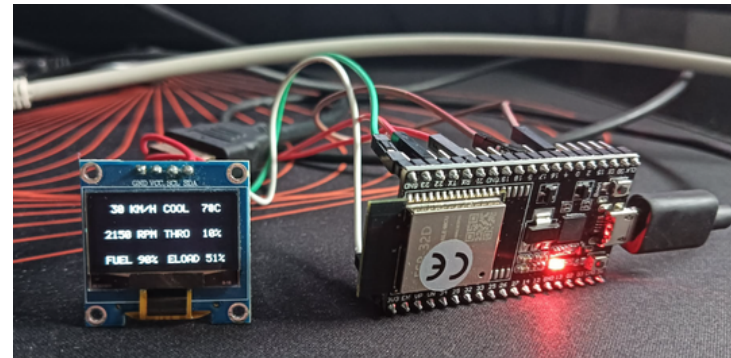
Total Cost ~ 120 zł / 30 €

PHASE OF DEVELOPMENT

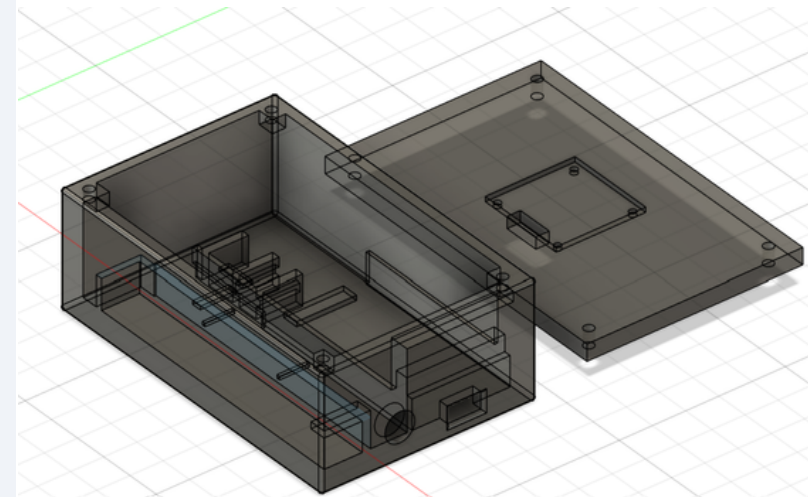
Assembly of Hardware



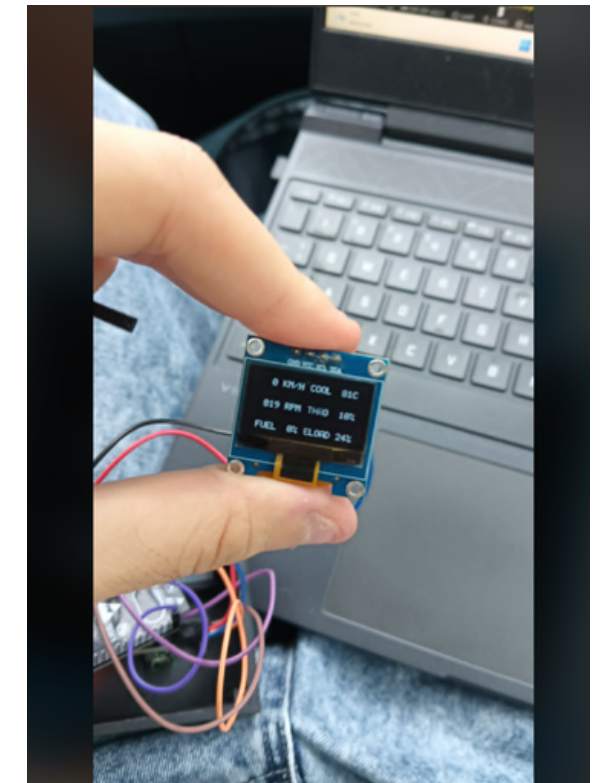
Simulation of Mock Frames in Visual Studio Code

[illegible]

Designing and Printing 3D case

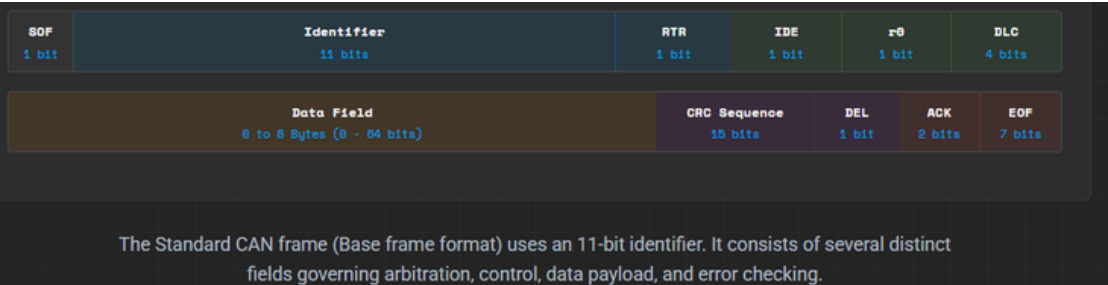


Test Performed on Real Car



SOFTWARE

CAN Frame Explained –

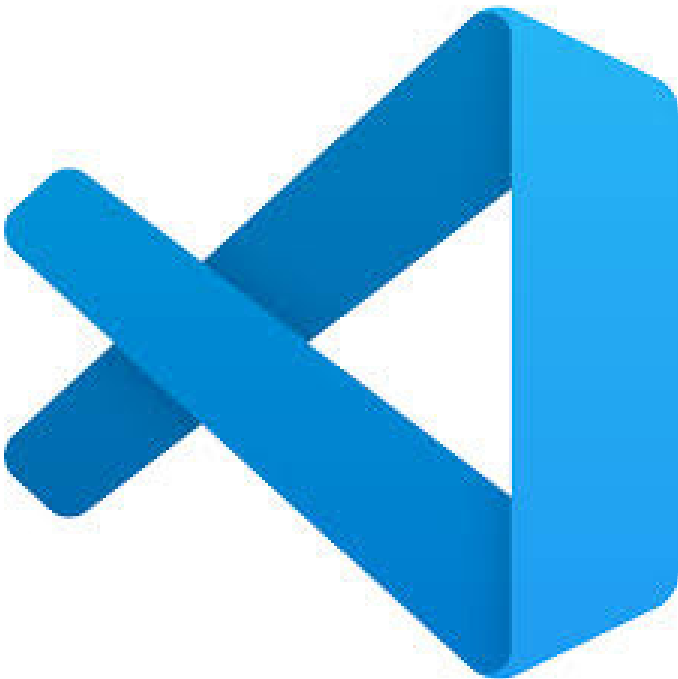


Mock Frames:

```
I (91753) RX: Ramka otrzymana ID: 0x7E8
I (91753) TWAI/CAN MOCK_test: Ramka CAN z Paramterami Odebarana 2300, Fuel Level: 49%, Speed: 155 KM/H, Coolant Temp: 90, Throttle Pos: 6, Engine Load: 97
I (91853) RX: Ramka otrzymana ID: 0x7E8
I (91853) TWAI/CAN MOCK_test: Ramka CAN z Paramterami Odebarana 2300, Fuel Level: 49%, Speed: 155 KM/H, Coolant Temp: 90, Throttle Pos: 6, Engine Load: 97
I (91953) RX: Ramka otrzymana ID: 0x7E8
I (91953) TWAI/CAN MOCK_test: Ramka CAN z Paramterami Odebarana 2300, Fuel Level: 49%, Speed: 155 KM/H, Coolant Temp: 90, Throttle Pos: 12, Engine Load: 97
I (92053) RX: Ramka otrzymana ID: 0x7E8
I (92053) TWAI/CAN MOCK_test: Ramka CAN z Paramterami Odebarana 2300, Fuel Level: 49%, Speed: 155 KM/H, Coolant Temp: 90, Throttle Pos: 12, Engine Load: 100
```

Real CAR ECU Frames:

```
(53051) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 11 18 00 00 00 00
(53221) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 11 18 00 00 00 00
(53221) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 04 3D 00 00 00 00
(53391) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 04 3D 00 00 00 00
(53391) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 04 41 0C 0C D8 00 00 00
(53731) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 04 41 0C 0C D8 00 00 00
(53731) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 00 00 00 00 00 00
(53901) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 00 00 00 00 00 00
(53901) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 05 78 00 00 00 00
(54071) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 05 78 00 00 00 00
(54071) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 11 18 00 00 00 00
(54241) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 11 18 00 00 00 00
(54241) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 04 3F 00 00 00 00
(54411) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 04 3F 00 00 00 00
(54411) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 04 41 0C 0C D8 00 00 00
(54751) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 04 41 0C 0C D8 00 00 00
(54751) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 00 00 00 00 00 00
(54921) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 00 00 00 00 00 00
(54921) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 05 79 00 00 00 00
(55091) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 05 79 00 00 00 00
(55091) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 11 18 00 00 00 00
(55261) RAW_RX: Odebrano ID: 0x7E8 | DLC: 8 | Data: 03 41 11 18 00 00 00 00
```



ISSUES and TROUBLESHOOTING

HARDWARE TRAPS

OBD - 2 Cable Mismatch

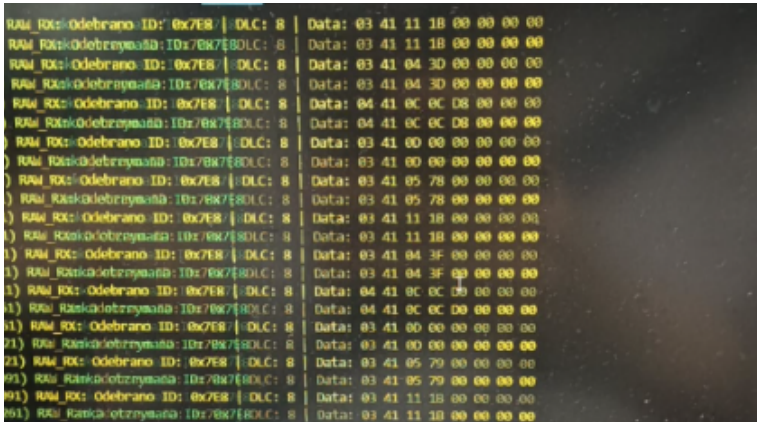


Wiriring

TX → TX
RX → RX

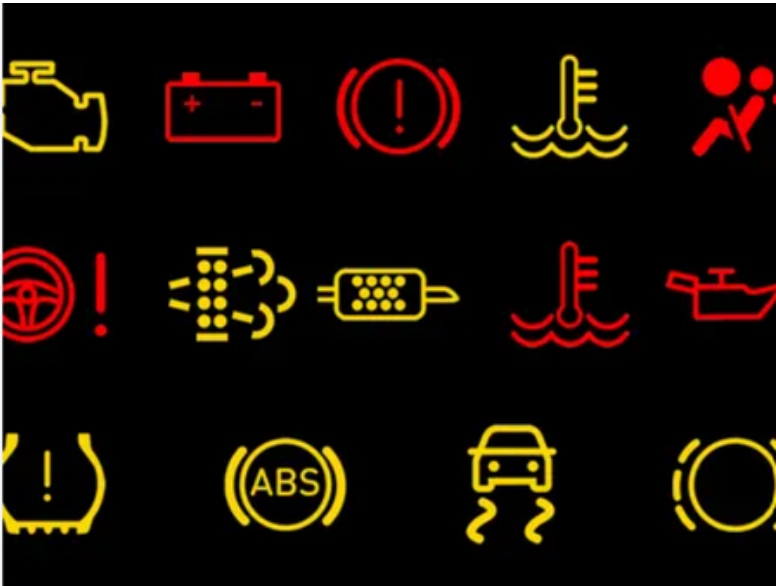
SOFTWARE

The Queue Overflow



CAR ERRORS

The Gateway Defense
and Warnings



Future PLANS

1. Multi-Protocol Expansion

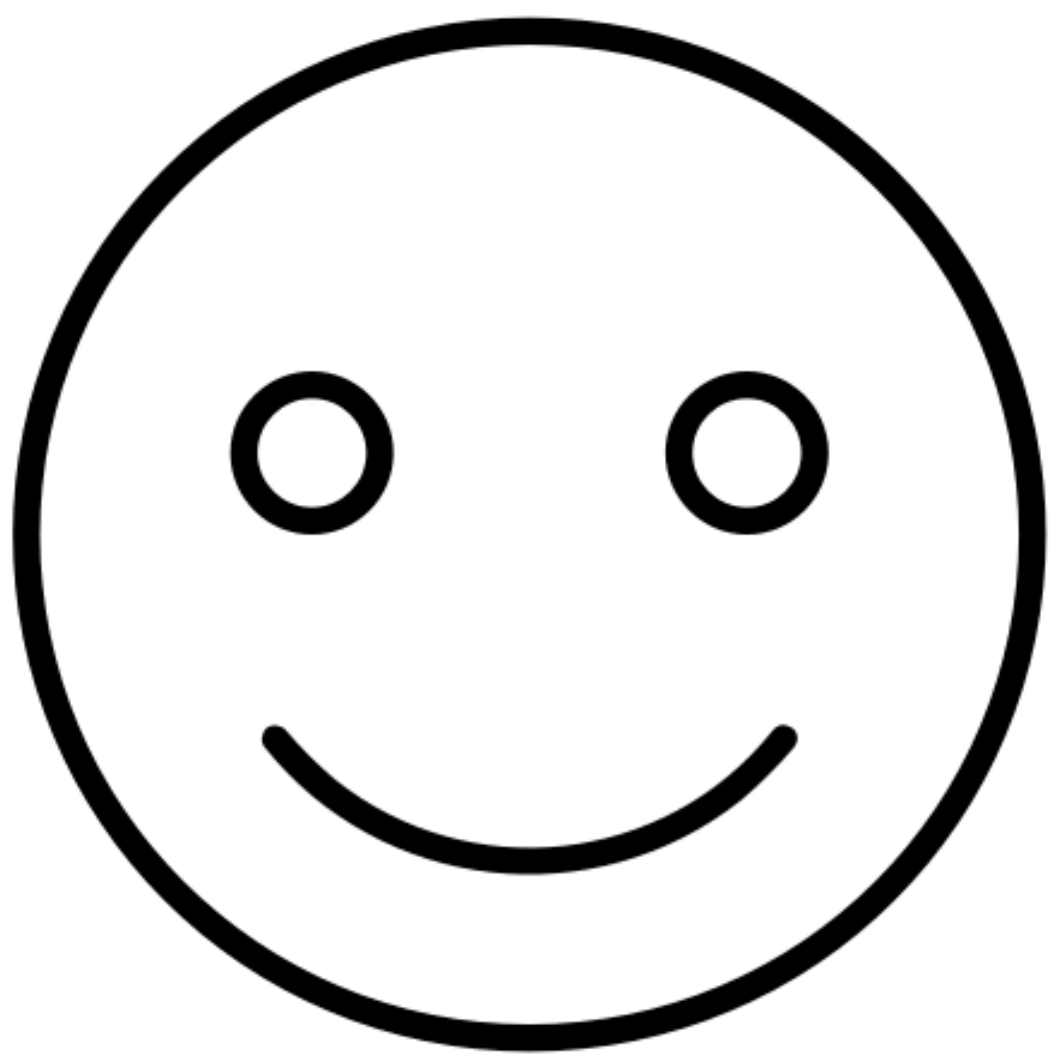
Add and Integarte another vehicle protocols like LIN, CAN FD , K-LINE ,

2. Wireless Telemetry

The ultimate goal is to broadcast this live telemetry to a smartphone dashboard or a cloud database for live analysis

3. Custom PCB

4. More TESTS!



Thank you for your
attention